

2026

Retrieval-Augmented Academic Assistant with Local RAG Implementation

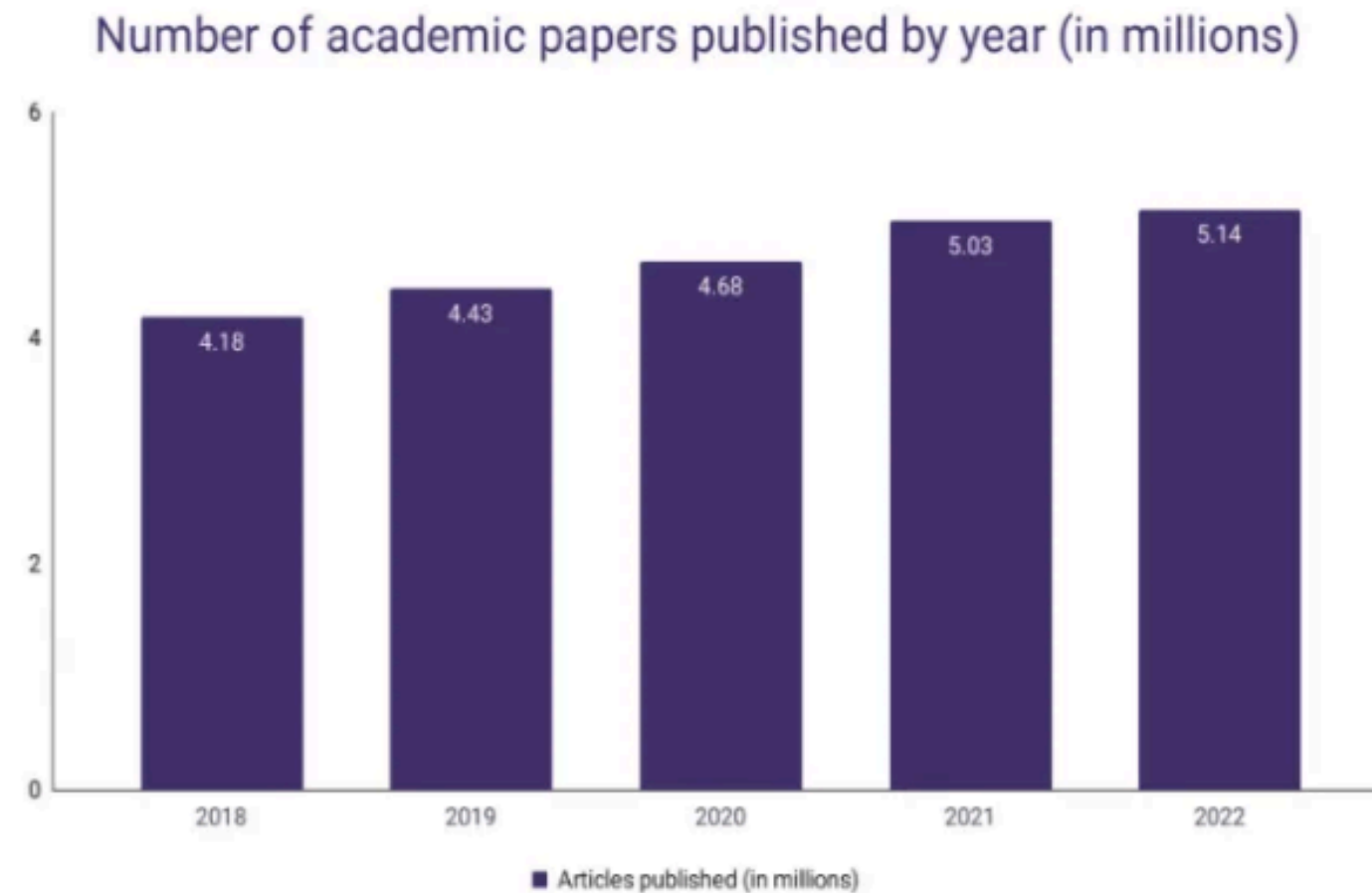
Master in Data Science Research Report
Faculty of Computer Science & Information Technology,
Universiti Malaya

Supervisor: Dr. LIEW WEI SHIUNG

Presented by: LIEW YEONG CHUAN



The Academic Information Overload Problem



Curcic, D. (2023). *Number of Academic Papers Published Per Year – WordsRated*. Wordsrated.
<https://wordrated.com/number-of-academic-papers-published-per-year/>

Over 5.14 Million Papers/Year

Academic publications grow at ~5.6% annually, outpacing traditional search systems (Hanson et al., 2024).

Vocabulary Mismatch

Sparse retrieval (BM25, TF-IDF) struggles with semantic relationships — different terms for the same concept reduce effectiveness.

LLM Hallucinations

LLMs alone produce outdated or hallucinated responses for domain-specific questions, making them unreliable in academic settings.

Problem Statement & Literature Review

→ **Domain-Specific Underexploration in Academic QA**

While RAG is heavily researched in healthcare and legal domains, its application to academic literature with highly specialized terminology remains underexplored

→ **Constraints of Cloud-Dependent Pipelines**

Cloud-dependent pipelines incur high operational costs, limit offline customization, and expose confidential or unpublished research to privacy risks

→ **Subjectivity and Poor Scalability of Human Evaluation**

Traditional reliance on human evaluation is subjective, slow, and non-reproducible, highlighting the lack of standardized, automated evaluation metrics

□ Therefore, there is a need for a locally deployed Retrieval-Augmented Generation pipeline for academic question answering. The pipeline needs to be constructed by systematically evaluating different retrieval and generator models using automatic evaluation metrics.

Research Questions & Objectives

Research Questions

RQ1: What technical framework is needed for a local RAG system for academic articles?

RQ2: Which retrieval and generator models perform best for local academic question answering?

RQ3: How to evaluate RAG components without human evaluation?

RQ4: How does the optimized pipeline compare to baseline via RAGAS?

Research Objectives

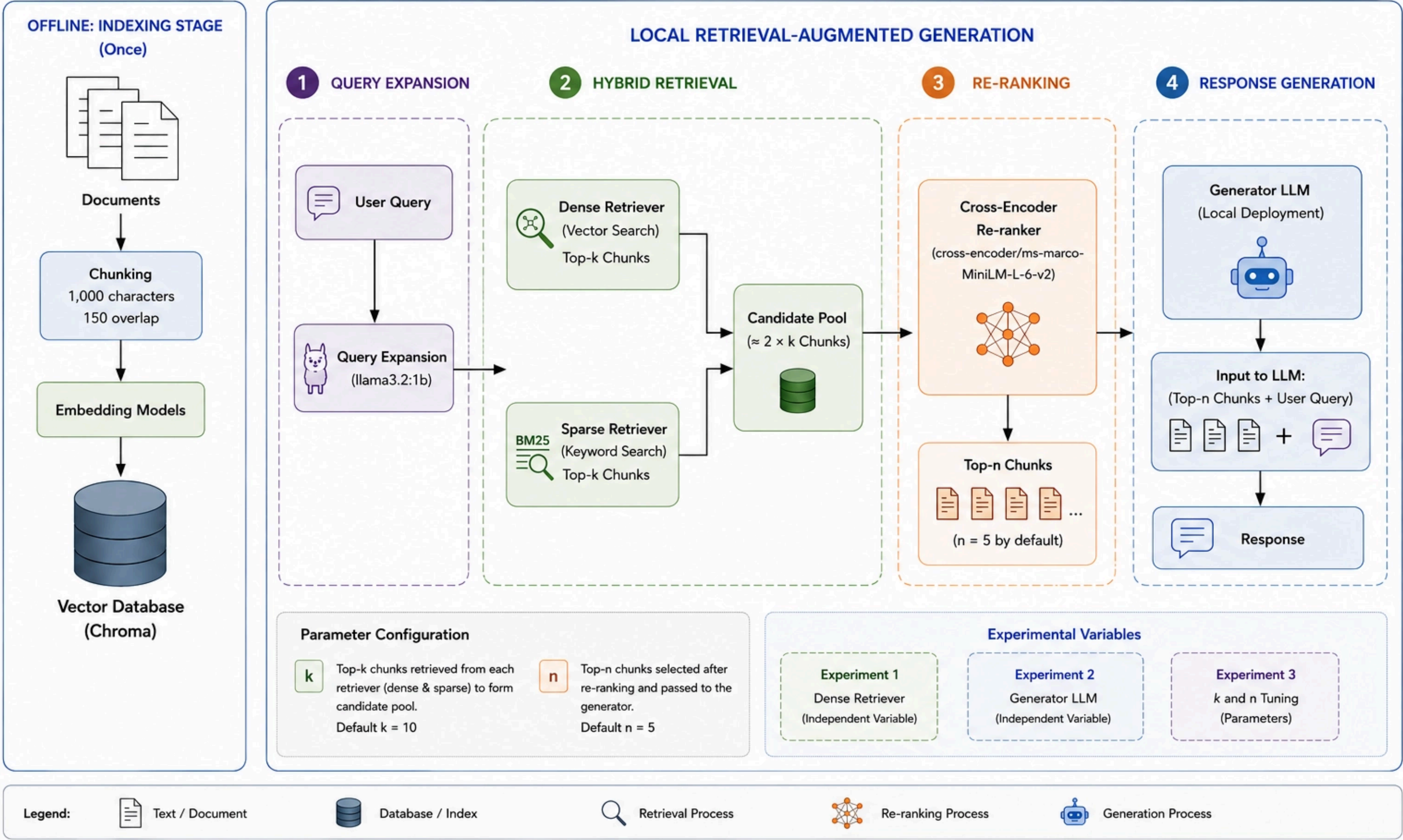
RO1: Design a local RAG pipeline for academic QA using local LLMs and vector databases.

RO2: Identify optimal retriever and generator models via automatic evaluation metrics.

RO3: Evaluate baseline vs. optimized pipelines using RAGAS metrics.

- ❏ **Scope:** English-language articles only; open-weight models under 8B parameters.
- ❏ **Significance of the Study:** contributes to Sustainable Development Goal 4, Quality Education,

Proposed Retrieval-Augmented Generation (RAG) Pipeline



Dataset & Benchmark Questionset Design

Dataset

Our corpus utilizes **Cornell University's public arXiv dataset** from Kaggle, employing the **abstract** and **summary fields** as the primary target knowledge source. It is curated via stratified random sampling to span five core fields across four temporal year groups from 2015–2024.

- **Source:** Cornell University's arXiv Dataset (Kaggle)
- **Size:** 136,238 records structured into 20 sub-datasets
- **Fields:** ML, Statistics, Computer Vision, NLP, and AI

Benchmark Questionset

A robust benchmark of 400 unique, ground-truth-linked questions was developed, systematically assigning 20 questions to each of the 20 dataset clusters. These queries span multiple task complexities, enabling standardized, automated pipeline evaluation

- **Questions:** 400 unique, ground-truth-linked queries
- **Categories:**
 - **Factual Lookup & Summarization:** Direct factual retrieval and single-article summaries.
 - **Comparative & Synthesis:** Multi-document cross-referencing and multi-hop reasoning.
 - **Negative / Unanswerable:** Refusal testing on queries designed to have no answer.
- **Evaluation:** Recall@n, NLI Semantic Consistency, Refusal Rate, and RAGAS metrics

Experiment Design

Experiment 1: Retriever Selection

Evaluate four open-source embedding models under identical pipeline conditions to identify the optimal dense retriever

- **Metric: Recall@5**
- **Candidate Models:**
 - all-minilm:22m
 - granite-embedding:30m
 - nomic-embed-text:v1.5
 - qwen3-embedding:0.6b

Experiment 2: Generator Selection

Evaluate five compact, local open-weight language models under static, controlled retrieval contexts

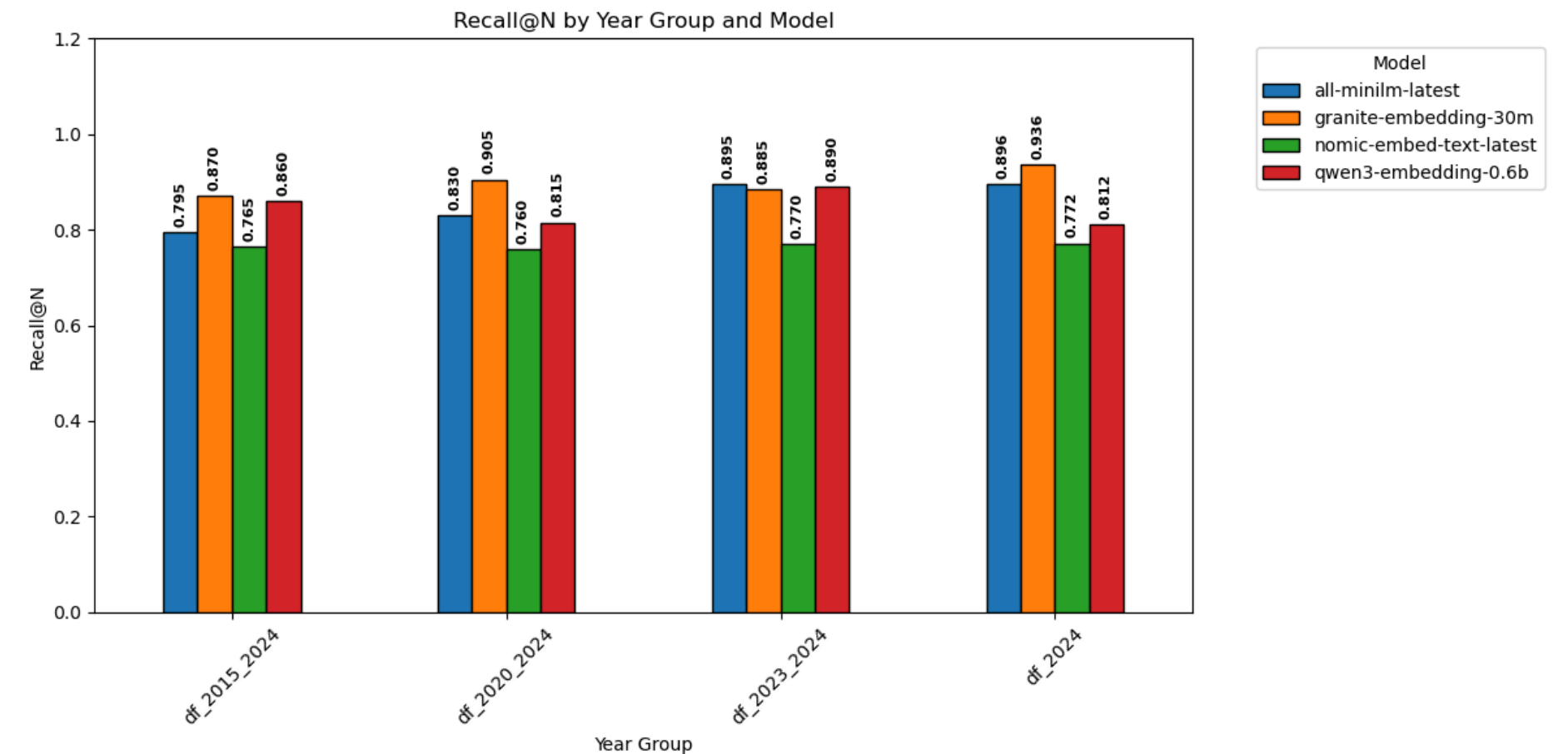
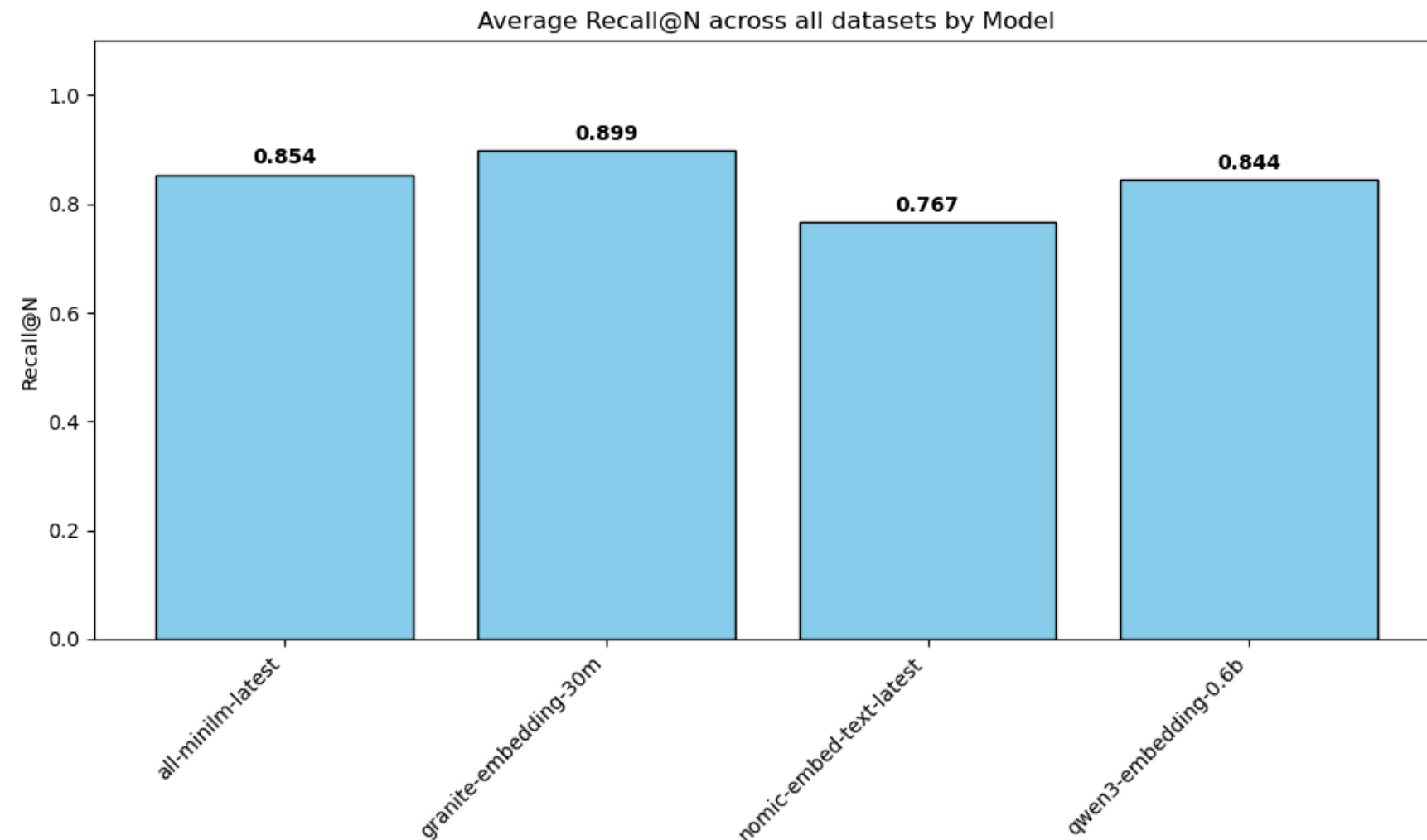
- **Metrics:**
 - NLI semantic consistency (under 3 Evaluator Models)
 - Refusal rate on unanswerable queries
- **Candidate Models:**
 - llama3.2:1b
 - llama3.2:3b
 - qwen2.5:3b
 - phi3.5:3.8b
 - gemma2:2b

Experiment 3: Pipeline Optimization

Conduct an end-to-end optimization of the fully integrated retriever-generator pipeline

- **Retriever Tuning:** Grid search for optimal k and n values
- **Generator Tuning:** Supervised Fine-Tuning using local QLoRA on NLI-filtered training datasets
- **Framework Evaluation:** Comparative assessment of baseline vs. optimized pipeline using RAGAS

Experiment 1 Results: Dense Retriever Selection



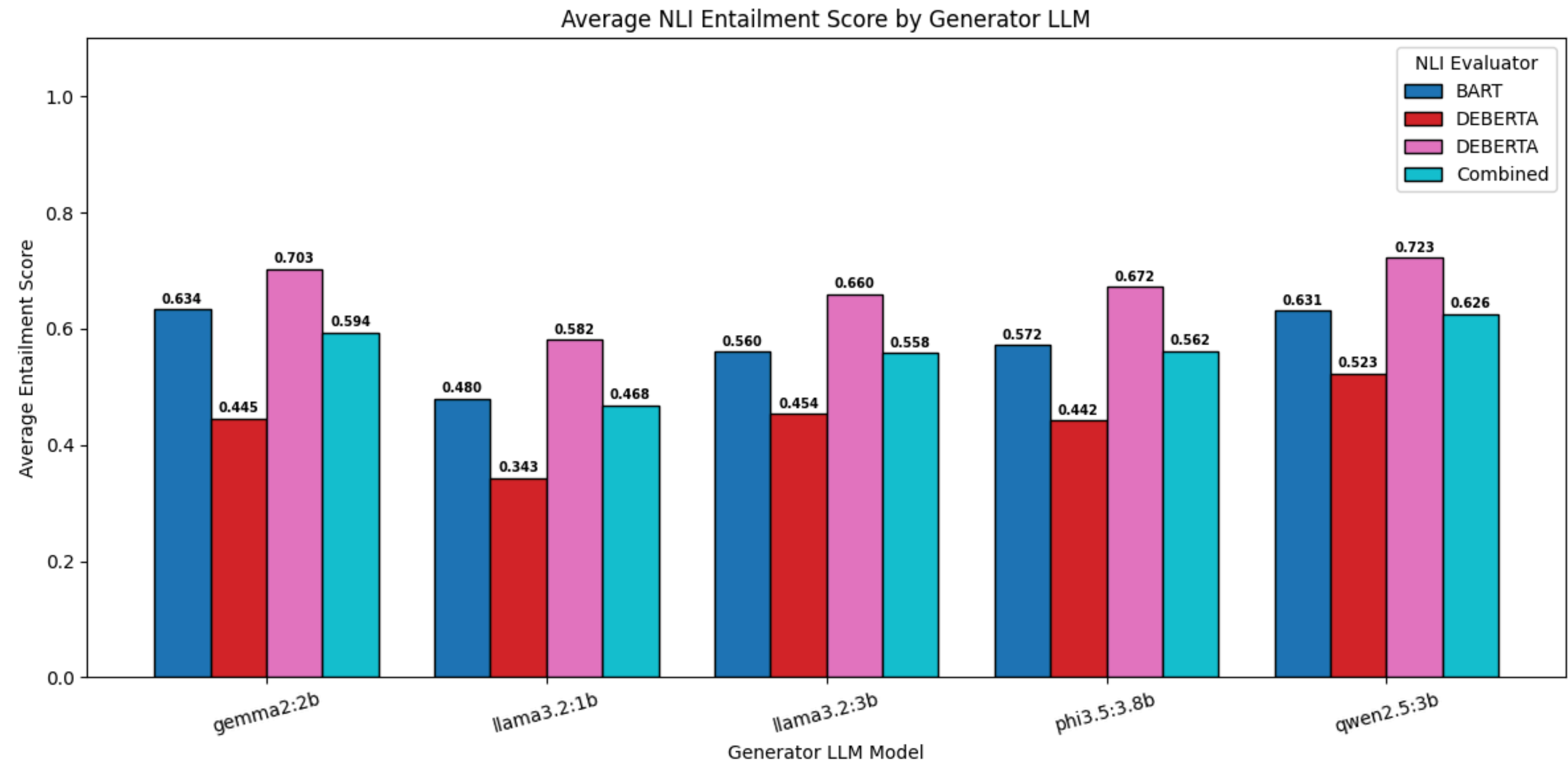
Key Findings

granite-embedding:30m achieved the highest Recall@5 at **89.9%**, consistently outperforming all models across every year group — peaking at **93.6%** on the 2024 dataset.

Notably, retrieval performance did not correlate with model size: the larger qwen3-embedding:0.6b (1024 dims) was outperformed by the smaller granite-embedding:30m (384 dims), suggesting training methodology matters more than scale.

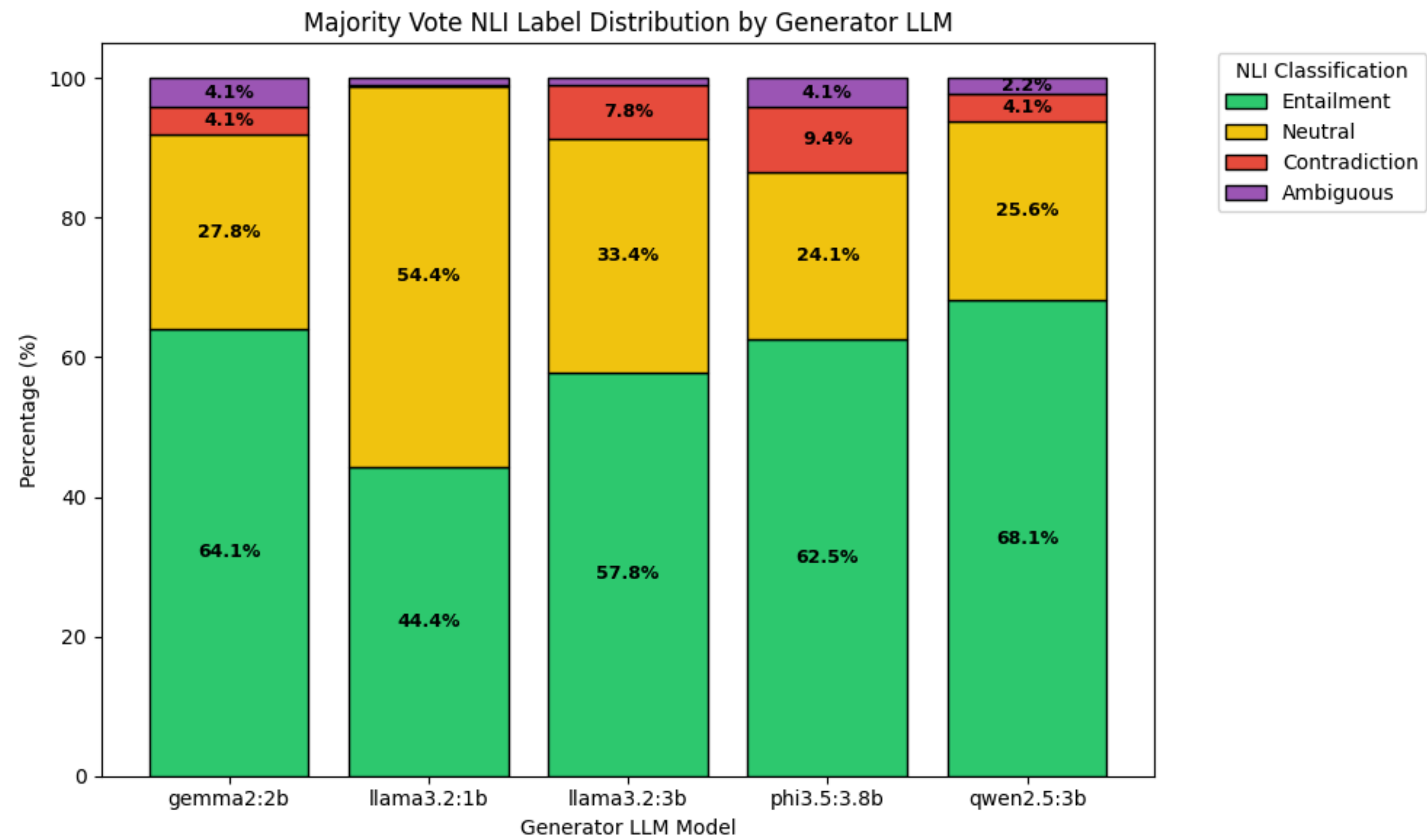
Experiment 2 Results: Generator Model Selection

Semantic Consistency (NLI)



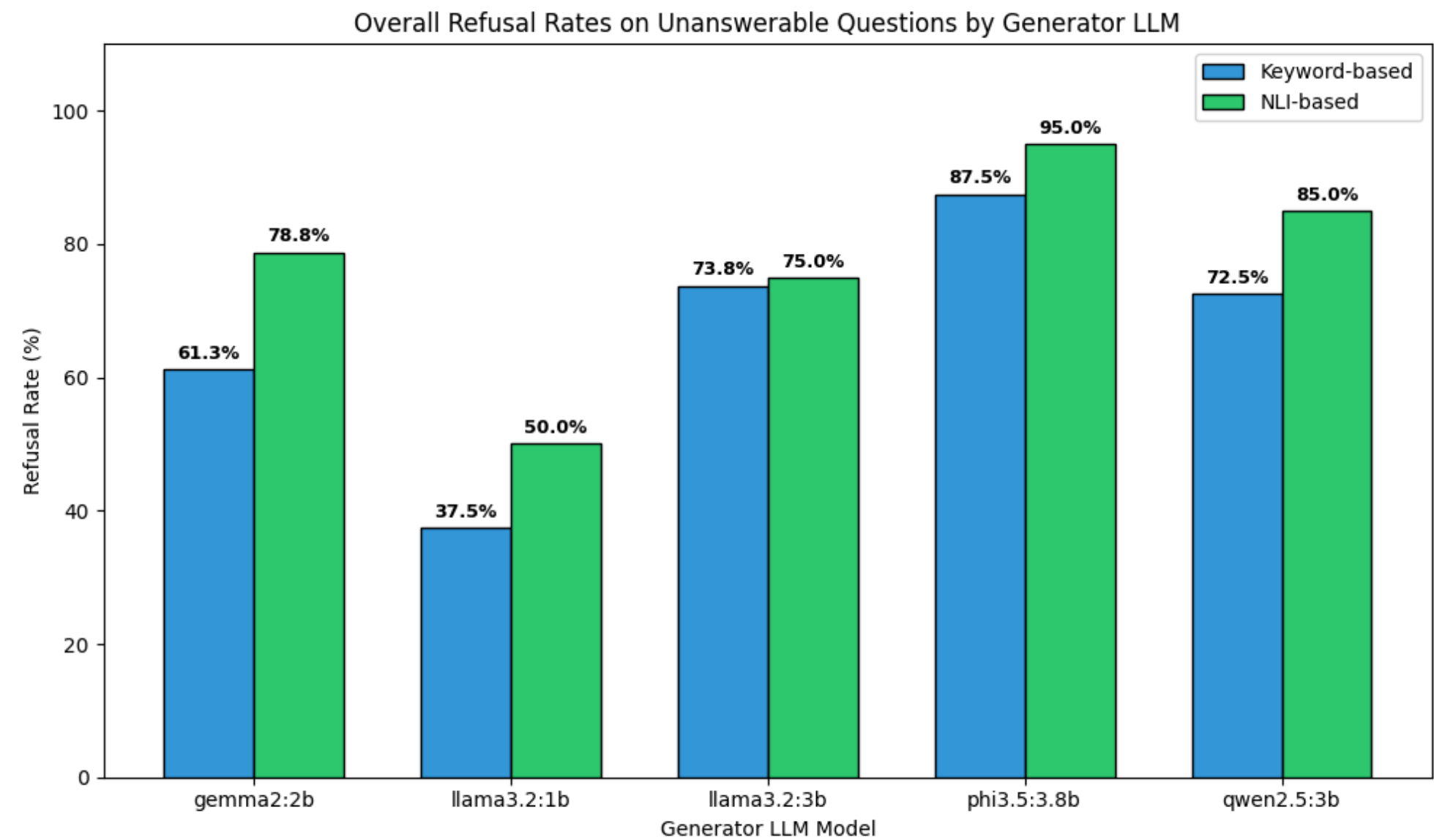
Experiment 2 Results: Generator Model Selection

Semantic Consistency (NLI)



Experiment 2 Results: Generator Model Selection

Refusal Rates



Final Selection: Qwen2.5:3B

qwen2.5:3b achieved the highest combined NLI entailment score (**0.626**) and the highest majority-vote entailment rate (**68.1%**), indicating the strongest semantic alignment with source articles.

It also demonstrated strong refusal capability (72.5% keyword-based, 85.0% NLI-based), second only to phi3.5:3.8b. Model architecture and training quality proved more influential than parameter count alone.

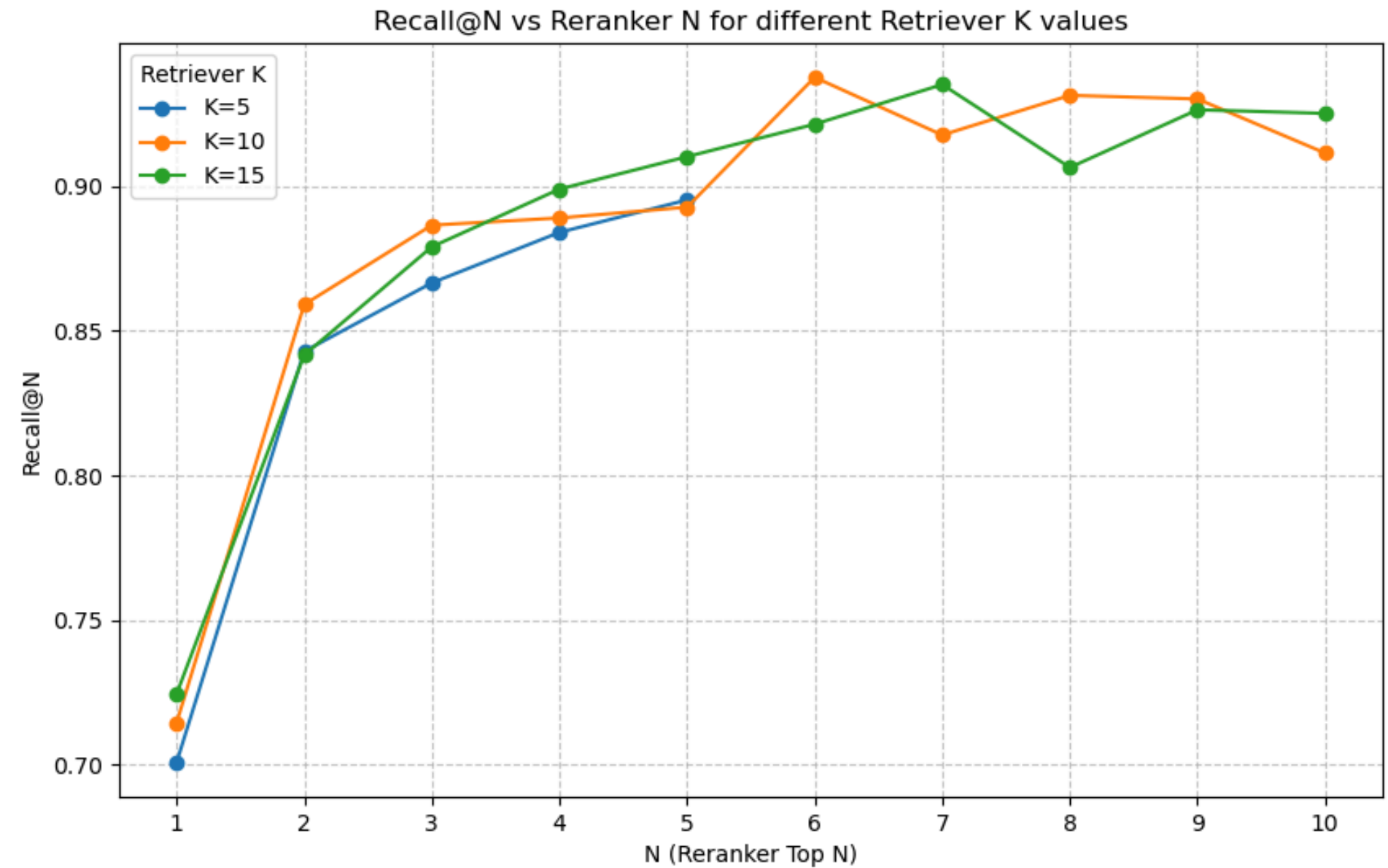
Experiment 3: Pipeline Optimization

Hyperparameter Tuning

Grid search over $k \in \{5, 10, 15\}$ and $n \in \{1-10\}$ identified $k = 10, n = 6$ as optimal — the elbow point of the Recall@n curve. Increasing k beyond 10 yielded diminishing returns while adding computational overhead.

QLoRA Fine-Tuning

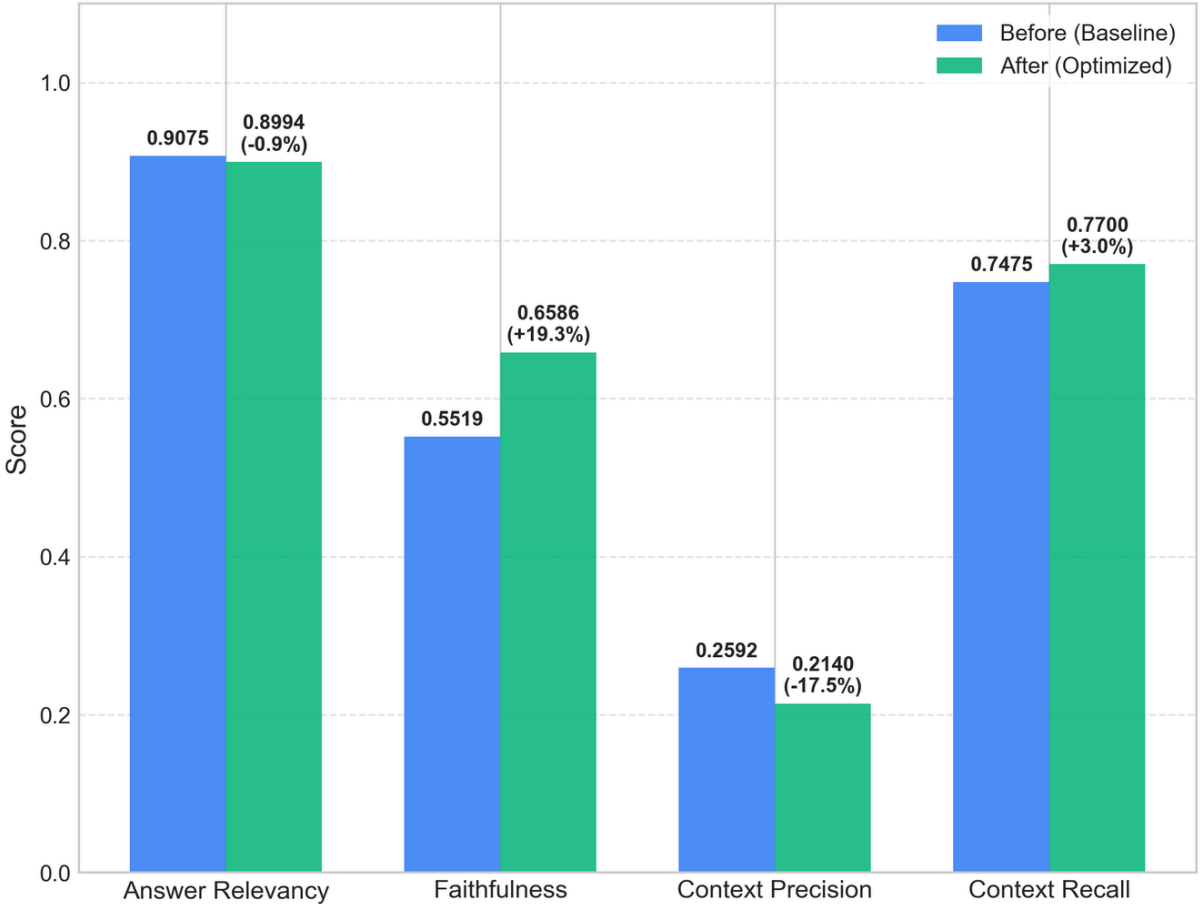
Qwen2.5-3B was fine-tuned using NLI-filtered high-quality responses, stratified sampling, and synthetic data augmentation (via llama3:8b). Only adapter parameters were updated, preserving base model capabilities.



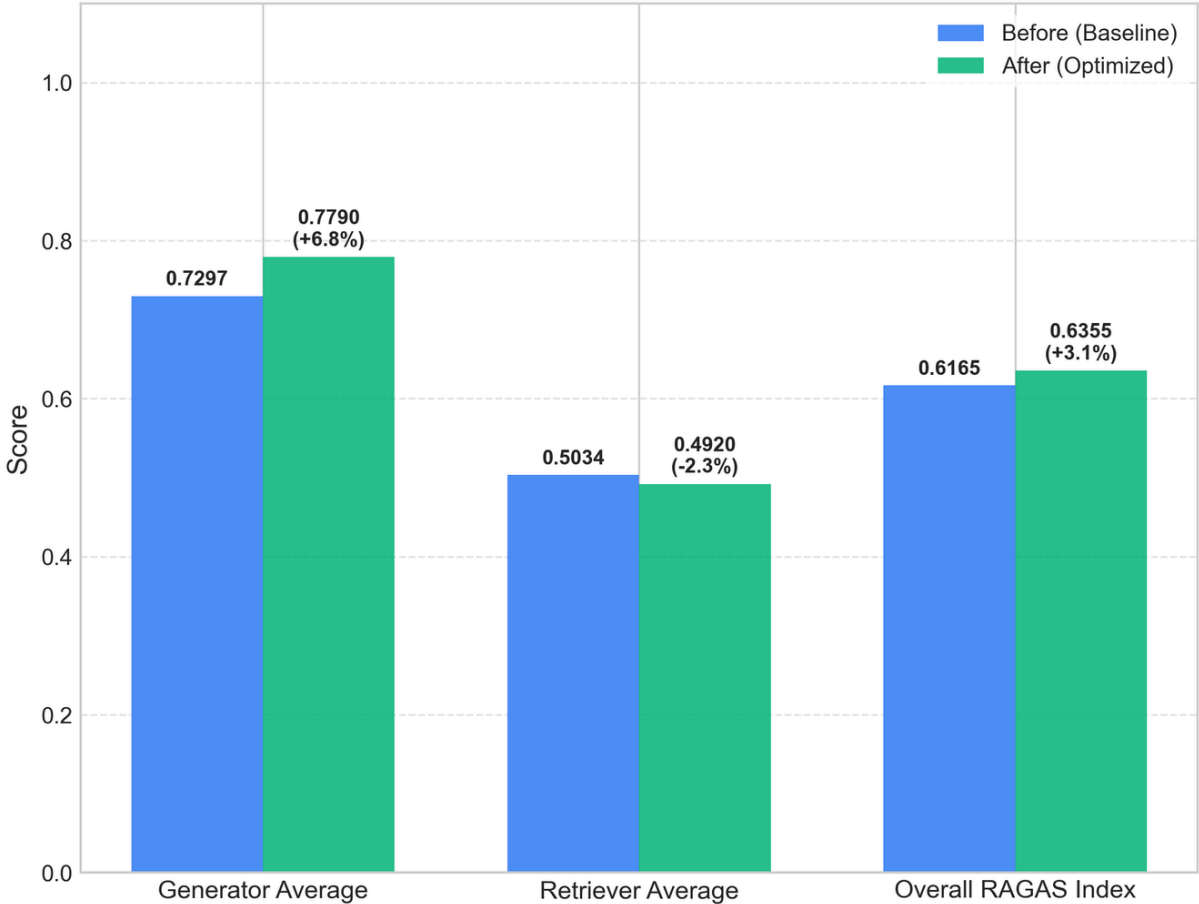
RAGAS Evaluation: Before vs. After Optimization

RAGAS Retrieval-Augmented Generation Evaluation Report

Core RAGAS Metrics Comparison



Aggregated Performance & Index



Key Improvements

+19.3% Faithfulness

The most critical gain — responses are far more grounded in retrieved evidence.

+3.0% Context Recall

Better retrieval coverage ensures more relevant documents reach the generator.

+3.1% RAGAS Index

Overall index improved from 0.6165 to 0.6355, confirming pipeline effectiveness.

📌 Context Precision decreased (−17.5%) as a trade-off for higher Recall — an intentional design choice prioritizing evidence coverage.

Conclusion & Contributions

→ RO1 Achieved: Local RAG Framework Developed

A fully reproducible, locally deployed RAG pipeline for academic QA — integrating query expansion, hybrid retrieval, cross-encoder re-ranking, and a fine-tuned local LLM.

→ RO2 Achieved: Optimal Components Identified

granite-embedding:30m (Retriever) and **Qwen2.5:3B** (Generator) selected via automatic evaluation — no human annotation required.

→ RO3 Achieved: Pipeline Optimized & Evaluated

Optimization achieved a **19.3% relative improvement in Faithfulness** and a higher overall RAGAS Index, demonstrating that lightweight open-weight models can effectively support academic QA.

Current Limitations:

- Restricted to English-language materials
- The knowledge base is built using abstract summaries rather than full-text papers

Future Works:

- Extend support to multilingual academic datasets
- Incorporate full-text research articles as knowledge sources
- Implement adaptive retrieval, dynamic context selection, and advanced re-ranking